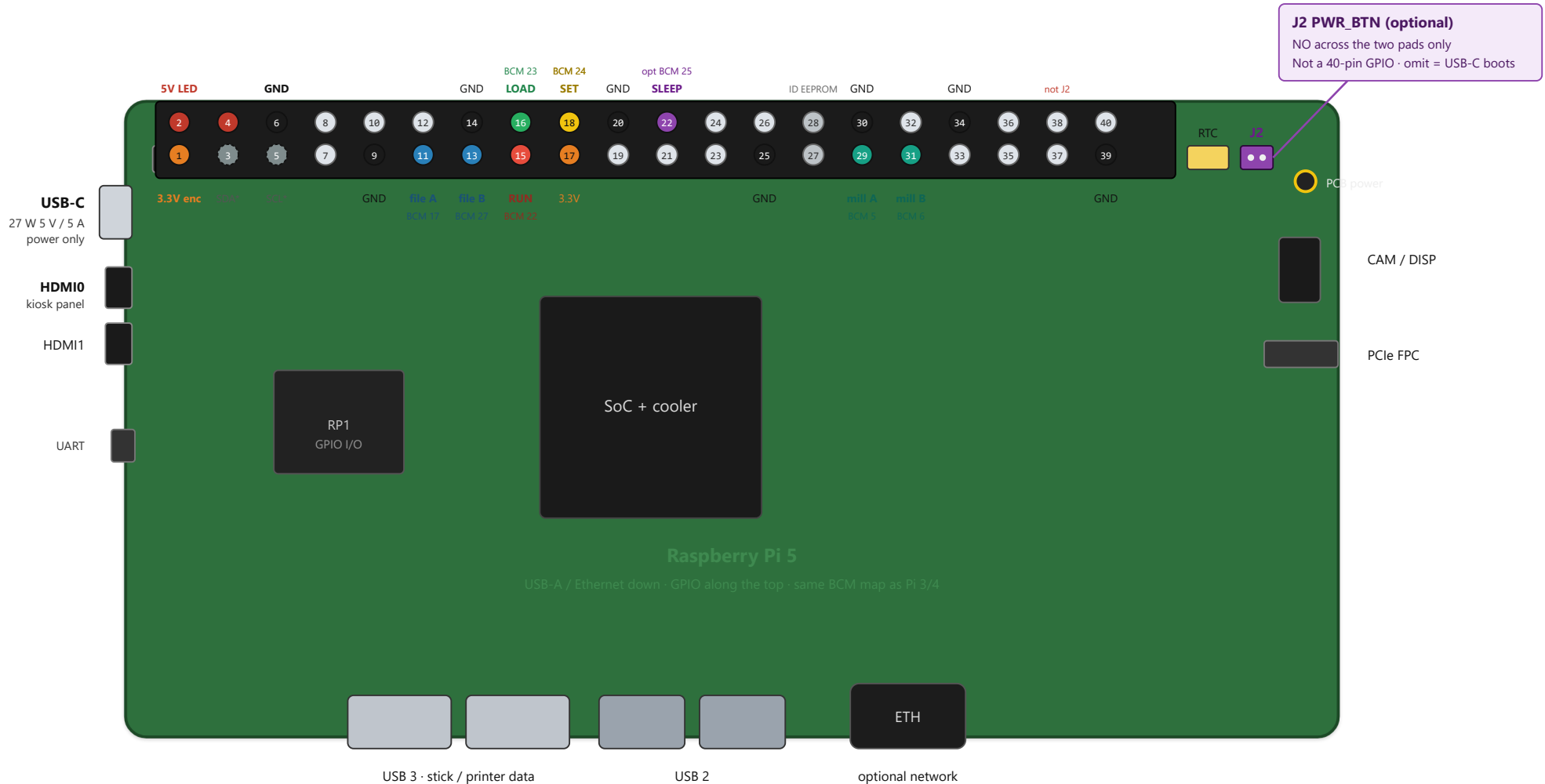


fh6parse 1.4.0 — Raspberry Pi 5 kiosk wiring

Hardware sheet only. Software / OS / UPDATE: packaging/LINUX-KIOSK.md. GPIO is **3.3 V**. BCM numbers, not header pin numbers. Optional parts may stay unwired.

40-pin GPIO · pin 1 at USB-C end · outer row = even (board edge) · inner row = odd (SoC)



3.3 V (encoders) 5 V (LED rings only) GND (common) File EC11 Mill EC11 LOAD green SET yellow RUN red Sleep (optional) I²C1 reserved* Unused

* Pins 3 / 5 (BCM 2 / 3, I²C1 SDA / SCL) are left free for a future Qwiic keypad. Do not steal them for F2 or extra buttons. Grey dashed = reserved, not required. Pin 38 (BCM 20) is **not** the power button.

1. Front panel (as you face the kiosk)



Buttons under the screen, left → right



header 16
BCM 23



header 18
BCM 24



header 15
BCM 22



header 22
BCM 25

Leave SLEEP unwired: pin sits on pull-up, never
POWER is J2, not this GPIO.

2. DFRobot Fermion EC11 (SEN0235)

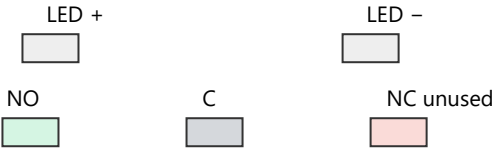
Two knobs, same module. 3.3–5 V — power from **3.3 V only**. Shaft click **C unused**.

EC11 silk	File knob	Mill knob
VCC	pin 1 (3.3 V)	pin 1 or 17 (3.3 V)
GND	pin 6 (GND)	pin 30 (GND)
A (CLK)	pin 11 · BCM 17	pin 29 · BCM 5
B (DT)	pin 13 · BCM 27	pin 31 · BCM 6
C (push)	not connected	

Wrong direction → settings **Reverse** (or swap A/B). 20 pulses/turn; start **Ticks per tooth = 2** if one click skips two names.

3. 16 mm 5-pin vandal (LOAD / SET / RUN / optional SLEEP)

Switch and ring LED are separate. Meter the tabs — clone silk varies. Rear view (typical):



Tab	To Pi
NO	GPIO (pull-up). Press = GND
C	common GND (pin 6 / 9 / 14 / 20 / 30)
NC	leave open
LED +	5 V pin 2 (or 4). Not 3.3 V
LED -	GND

4. Optional J2 power (not GPIO)

Purple 2-pad header next to the RTC battery, GPIO pin-40 corner. Same switch as the PCB button.

Do	Do not
Momentary NO across the two J2 pads	Tie J2 to BCM 25, BCM 20, or GPIO GND

5. Power and USB (not GPIO)

Port	Use
USB-C (Pi)	Official 27 W PSU. Power only
micro-HDMI0	Kiosk LCD (port next to USB-C)
USB-A	Stick, and P047 data (printer has its own mains PSU)
Ethernet	Optional (git / UPDATE / company STEP share)

Leave J2 open if unused — USB-C still boots	Expect Pi 4 WAKE_ON_GPIO on GPIO3
Keep the PCB button reachable	Put LED rings on J2 · set WAIT_FOR_POWER_BUTTON unless J2 is fitted

Do not power the P047 from Pi USB. Rings stay on 5 V after halt unless you switch that 5 V.

6. Default map (header pin → BCM → function)

Header	BCM	Function	Required
1	—	3.3 V · encoder VCC	yes if knobs fitted
2 / 4	—	5 V · vandal LED +	rings only
3	2	I ² C1 SDA — reserved, leave open	no
5	3	I ² C1 SCL — reserved, leave open	no
6, 9, 14, 20, 30	—	GND · common for knobs, switches, LED –	yes
11	17	File encoder A (CLK)	yes
13	27	File encoder B (DT)	yes
15	22	RUN red NO	yes
16	23	LOAD green NO	yes
17	—	3.3 V spare (same rail as pin 1)	no
18	24	SET yellow NO	yes
22	25	SLEEP / wake NO	no — unconnected is fine
29	5	Mill encoder A (CLK)	yes
31	6	Mill encoder B (DT)	yes
J2 pads	—	Panel POWER (parallel with PCB button)	no — USB-C still boots

Change BCM numbers in kiosk settings (**F2**) or `/etc/fh6parse-kiosk.ini`. Overlay: `~/config/fh6parse/ui.ini`.

7. Common ground

One GND net for encoders, every C tab, every LED –, and (if fitted) a Qwiic keypad. Tie them to any GND header pin listed above. Do not run a separate “button ground” to J2.

8. Identify vandal tabs with a meter

1. Diode pair = LED + / – (one way ~1.8–3 V).
2. Unpressed: C–NC beeps, C–NO open.
3. Pressed: C–NO beeps. The tab that beeps in both tests is **C**.

Do not

- Put 5 V on A / B / C / NO / NC or any BCM pin
- Power EC11 VCC from pin 2 (5 V) unless a shifter is on A/B
- Wire J2 to GPIO 25 or GPIO 20
- Use `Rpi.GPIO` on Pi 5 (RP1 needs `Igpio`)
- Set `WAIT_FOR_POWER_BUTTON=1` without a J2 (or reachable PCB) button

Without optional parts the kiosk still runs: idle blanks the panel; knobs / USB / keyboard wake it; USB-C plug-in boots the Pi. `gpiozero` pull-ups keep unused BCM 25 high.